

FINANCIAL MANAGEMENT

I.M. Pandey - 12th Edition

Exercise Solutions with Detailed Calculations

Chapters 8, 10 and 12

CHAPTER 8: CAPITAL BUDGETING - NPV, IRR, ARR

Question 2: NPV and IRR of Machine Investment

A machine costs Rs.1,00,000 and provides annual net cash inflow of Rs.30,000 for 6 years. Cost of capital is 15%.

Given Data:

Parameter	Value
Initial Investment	Rs.1,00,000
Annual Net Cash Inflow	Rs.30,000
Life	6 years
Cost of Capital	15%

(a) NPV Calculation:

NPV = Sum of $[\text{Cash Flow} / (1 + r)^t]$ for $t = 0$ to 6

Year	Cash Flow (Rs.)	PV Factor @15%	Present Value (Rs.)
0	-1,00,000	1.0000	-1,00,000.00
1	30,000	0.8696	26,086.96
2	30,000	0.7561	22,684.31
3	30,000	0.6575	19,725.49
4	30,000	0.5718	17,152.60
5	30,000	0.4972	14,915.30
6	30,000	0.4323	12,969.83
		NPV =	13,534.48

(b) IRR Calculation:

The IRR is the discount rate at which NPV = 0. Using the annuity relationship:

$$1,00,000 = 30,000 \times \text{PVIFA}(r, 6) \Rightarrow \text{PVIFA}(r, 6) = 3.3333$$

From annuity tables, at 19% PVIFA = 3.4098 and at 20% PVIFA = 3.3255.

By interpolation: $\text{IRR} = 19\% + (3.4098 - 3.3333)/(3.4098 - 3.3255) \times 1\% = 19\% + 0.91\% = \mathbf{19.91\%}$

(c) Decision:

The machine SHOULD be purchased. NPV = Rs.13,534.48 (positive) and IRR = 19.91% > Cost of Capital 15%. Both criteria support acceptance.

Question 6: Refrigerator Instalment - Distributor's Rate of Return

Refrigerator costs Rs.10,000. Distributor charges Rs.3,700/year for 4 years (total Rs.14,800). Stated rate is 12%. Opportunity cost is 14%.

(a) Actual Rate of Return (IRR) earned by Distributor:

The distributor's cash flows: Outflow Rs.10,000 at Year 0, Inflows Rs.3,700 per year for 4 years.

$$10,000 = 3,700 \times \text{PVIFA}(r, 4) \Rightarrow \text{PVIFA}(r, 4) = 2.7027$$

From tables: at 17% PVIFA = 2.7432 and at 18% PVIFA = 2.6901.

$\text{IRR} = 17\% + (2.7432 - 2.7027)/(2.7432 - 2.6901) \times 1\% = 17\% + 0.76\% = \mathbf{17.76\%}$

(b) NPV at 14% Opportunity Cost:

Year	Cash Flow (Rs.)	PV Factor @14%	PV (Rs.)
0	-10,000	1.0000	-10,000.00

Year	Cash Flow (Rs.)	PV Factor @14%	PV (Rs.)
1	3,700	0.8772	3,245.61
2	3,700	0.7695	2,847.03
3	3,700	0.6750	2,497.39
4	3,700	0.5921	2,190.70
		NPV =	780.74

Decision:

The NPV at 14% is Rs.780.74 (positive from the distributor's perspective). The effective interest rate the distributor is charging is 17.76%, NOT 12% as stated. Since the buyer's effective borrowing rate (17.76%) > opportunity cost (14%), **the buyer should NOT accept the offer** if alternative financing at 14% or lower is available.

Question 12: NPV Profiles for Projects X and Y

Given Cash Flows:

Year	Project X	Project Y
0	-750	-750
1	+350	+250
2	+350	+250
3	+159	+460

(a) NPV at Various Discount Rates:

Discount Rate	NPV - Project X	NPV - Project Y	NPV (Y-X)
0.00%	109.00	210.00	101.00
5.00%	38.14	112.22	74.07
8.00%	0.36	60.98	60.62
10.00%	-23.10	29.49	52.59
12.00%	-45.31	-0.07	45.24
20.00%	-123.26	-101.85	21.41

(b) IRR from the table above:

IRR of Project X = 8.03% (NPV changes sign between 8% and 10%)

IRR of Project Y = 12.00% (NPV changes sign between 10% and 12%)

(c) When should Project X be accepted?

Project Y dominates Project X at all practical discount rates. Y has a higher IRR (12% vs 8%) and higher NPV at every discount rate tested. The incremental NPV (Y - X) is always positive at feasible rates. **Project X should only be accepted if Project Y is unavailable and the cost of capital is below 8%.**

(d) Incremental Analysis (Y - X):

Year	Incremental CF (Y-X)
0	0
1	-100
2	-100
3	301

Crossover IRR (where NPV of Y-X = 0) = **30.55%**. Since this is far above both projects' IRRs, Y is preferred over X whenever either project has a positive NPV.

Question 16: Accounting Rate of Return (ARR)

Given Income Statement (Rs.'000):

Particulars	C1	C2	C3
Cash Revenue	16	14	12
Cash Expenses	8	7	6
Gross Profit	8	7	6
Depreciation	4	4	4
Net Profit	4	3	2

Initial Investment = Rs.12,000

(a) ARR Calculation:

Average Net Profit = $(4 + 3 + 2) / 3 = \text{Rs.}3,000$

Average Investment = $12,000 / 2 = \text{Rs.}6,000$

ARR = Average Net Profit / Average Investment = $3,000 / 6,000 = 50.00\%$

(b) If Initial Investment = Rs.9,000 and Cash Expenses increase by Rs.1,000:

New Depreciation = $9,000 / 3 = \text{Rs.}3,000$ per year

Particulars	C1	C2	C3
Cash Revenue	16	14	12
Cash Expenses (+1)	9	8	7
Gross Profit	7	6	5
Depreciation	3	3	3
Net Profit	4	3	2

Average Net Profit = $(4 + 3 + 2) / 3 = \text{Rs.}3,000$

Average Investment = $9,000 / 2 = \text{Rs.}4,500$

ARR = $3,000 / 4,500 = 66.67\%$

NPV at 9% Cost of Capital (Rs.'000):

Year	Cash Flow	PV Factor @9%	PV (Rs.'000)
0	-9	1.0000	-9.000
1	7	0.9174	6.422
2	6	0.8417	5.050
3	5	0.7722	3.861
		NPV =	6.333

NPV = Rs.6,333 (positive). The project should be accepted.

CHAPTER 10: COST OF CAPITAL AND PROJECT ANALYSIS

Question 2: Mutually Exclusive Projects - Payback, IRR, NPV, PI

Cost of Capital = 10%, Tax Rate = 35%. Before-tax cash flows:

Year	Project X	Project Y
0	-20,000	-15,000
1	4,200	4,200
2	4,800	4,500
3	7,000	4,000
4	8,000	5,000
5	2,000	1,000

Depreciation: Project X = 20,000/5 = Rs.4,000/yr; Project Y = 15,000/5 = Rs.3,000/yr

After-Tax CF = BT CF x (1 - Tax) + Depreciation x Tax

Year	AT CF - X	AT CF - Y	PV@10%	PV-X	PV-Y	Cum X	Cum Y
0	-20,000	-15,000	1.0000	-20,000.00	-15,000.00	-20,000	-15,000
1	4,130	3,780	0.9091	3,754.55	3,436.36	-15,870	-11,220
2	4,520	3,975	0.8264	3,735.54	3,285.12	-11,350	-7,245
3	5,950	3,650	0.7513	4,470.32	2,742.30	-5,400	-3,595
4	6,600	4,300	0.6830	4,507.89	2,936.96	1,200	705
5	2,700	1,700	0.6209	1,676.49	1,055.57	3,900	2,405

NPV of Project X = Rs.-1,855.22

NPV of Project Y = Rs.-1,543.69

(a) Payback Period:

Project X: Cumulative AT CF turns positive between Year 3 (-5,400) and Year 4 (+1,200). Payback = 3 + 5,400/6,600 = **3.82 years**

Project Y: Cumulative AT CF turns positive between Year 3 (-3,595) and Year 4 (+705). Payback = 3 + 3,595/4,300 = **3.84 years**

(b) IRR:

IRR of Project X = 6.29% (below cost of capital)

IRR of Project Y = 5.59% (below cost of capital)

(c) NPV: Both NPVs are negative (X = Rs.-1,855; Y = Rs.-1,544).

(d) Profitability Index: $PI(X) = 0.91$; $PI(Y) = 0.90$

Decision:

Both projects should be rejected. Both have negative NPVs, IRRs below cost of capital, and $PI < 1$. If forced to choose, Project Y has a less negative NPV (Rs.-1,544 vs Rs.-1,855) and shorter payback, making it marginally less unfavourable.

Question 8: Ram Singh - Investment Alternatives

Ram Singh (age 45, retires at 58) has Rs.3,00,000 inheritance. Current salary Rs.36,000/year. Tax rate 30%.

ALTERNATIVE 1: Continue Job + Fixed Deposit at 15%

FD Maturity Value = $3,00,000 \times (1.15)^{13} = \text{Rs.18,45,836}$

Interest Earned = $18,45,836 - 3,00,000 = \text{Rs.15,45,836}$

Total Salary over 13 years = $36,000 \times 13 = \text{Rs.4,68,000}$

ALTERNATIVE 2: General Store

Parameter	Value
Initial Investment	Rs.2,40,000
Working Capital	Rs.60,000
Total Investment	Rs.3,00,000
Annual Receipts	Rs.3,90,000
Annual Out-of-Pocket Costs	Rs.3,00,000
Salvage Value (after 13 yrs)	Rs.50,000
Depreciation (SLM)	Rs.14,615/yr

Annual Income Calculation:

Particulars	Amount (Rs.)
Cash Profit (3,90,000 - 3,00,000)	90,000
Less: Depreciation	14,615
Taxable Income	75,385
Less: Tax @ 30%	22,615
Net Income after Tax	52,769
Add: Depreciation	14,615
After-Tax Cash Flow	67,385
Less: Salary Foregone	36,000
Net Incremental CF	31,385

NPV of Store (discounted at 15%):

Year	Cash Flow	PV Factor	PV (Rs.)
0	-3,00,000	1.0000	-3,00,000
1	67,385	0.8696	58,595.32
2	67,385	0.7561	50,952.45
3	67,385	0.6575	44,306.48
4	67,385	0.5718	38,527.37
5	67,385	0.4972	33,502.06
6	67,385	0.4323	29,132.23
7	67,385	0.3759	25,332.37
8	67,385	0.3269	22,028.15
9	67,385	0.2843	19,154.91
10	67,385	0.2472	16,656.45
11	67,385	0.2149	14,483.87
12	67,385	0.1869	12,594.67
13	177,385	0.1625	28,829.96
		NPV =	94,096.28

Decision:

The store is more profitable. NPV of the store at 15% = Rs.94,096.28 (positive). This means the store generates returns exceeding the 15% FD rate. Even after accounting for the salary foregone, the store creates substantial additional wealth compared to the FD alternative.

Question 11: Alpha Limited - Machine Replacement

New machine costs Rs.360 mn. Old machine exchange value Rs.40 mn. Required return 12%. Tax rate 50%.

Incremental Cost Savings (Rs. million/year):

Item	Old Machine	New Machine	Savings
Material Cost	240	195	45
Direct Labour	180	120	60
Variable Overheads	75	50	25
Indirect Labour	90	100	-10
Total Savings			120

Note: Allocated fixed overheads and depreciation are excluded as they are not incremental cash costs.

Net Investment = 360 - 40 = Rs.320 million

After-Tax Savings = 120 x (1 - 0.50) = Rs.60 mn/year

Incremental SLM Depreciation (approx) = (360-40)/10 - (20-6)/10 = 32 - 1.4 = Rs.30.6 mn

Tax Shield on Depreciation = 30.6 x 0.50 = Rs.15.3 mn

Annual Incremental ATCF = 60 + 15.3 = Rs.75.3 mn

Year	Inc. CF (Rs. mn)	PV Factor @12%	PV (Rs. mn)
0	-320.00	1.0000	-320.00
1	75.30	0.8929	67.23
2	75.30	0.7972	60.03
3	75.30	0.7118	53.60
4	75.30	0.6355	47.85
5	75.30	0.5674	42.73
6	75.30	0.5066	38.15
7	75.30	0.4523	34.06
8	75.30	0.4039	30.41
9	75.30	0.3606	27.15
10	109.30	0.3220	35.19
		NPV =	116.41

Part 1 Decision:

NPV = Rs.116.41 million (positive). The old machine should be replaced. The cost savings from the new machine more than justify the investment at 12% required return.

Part 2: External Purchase at Rs.40/unit

New machine variable cost/unit = (195 + 120 + 100 + 50) / 15 = Rs.31/unit

External purchase price = Rs.40/unit > Rs.31/unit (variable cost of new machine)

Since the external price (Rs.40) exceeds the variable manufacturing cost (Rs.31), the company should continue manufacturing internally with the new machine rather than outsourcing.

Question 18: Investment with Inflation Adjustment

Initial Cost = Rs.2,00,000; Life = 5 years; Annual Real Cash Inflow = Rs.70,000; Nominal Cost of Capital = 10%; Inflation = 5%

Part 1: Without Inflation (NPV at 10%)

Year	Cash Flow (Rs.)	PV Factor @10%	PV (Rs.)
0	-2,00,000	1.0000	-2,00,000.00
1	70,000	0.9091	63,636.36
2	70,000	0.8264	57,851.24
3	70,000	0.7513	52,592.04
4	70,000	0.6830	47,810.94
5	70,000	0.6209	43,464.49
		NPV =	65,355.07

NPV = Rs.65,355.07 (positive). Accept the project.

Part 2: With Inflation

Real Discount Rate = $(1 + 0.10)/(1 + 0.05) - 1 = 0.0476 = 4.76\%$

Method A: Discount Real CFs at Real Rate

Year	Real CF (Rs.)	PV Factor @4.76%	PV (Rs.)
0	-2,00,000	1.0000	-2,00,000.00
1	70,000	0.9545	66,818.18
2	70,000	0.9112	63,780.99
3	70,000	0.8697	60,881.86
4	70,000	0.8302	58,114.50
5	70,000	0.7925	55,472.93
		NPV =	105,068.46

Method B: Discount Nominal CFs at Nominal Rate

Year	Nominal CF (Rs.)	PV Factor @10%	PV (Rs.)
0	-2,00,000	1.0000	-2,00,000.00
1	73,500.00	0.9091	66,818.18
2	77,175.00	0.8264	63,780.99
3	81,033.75	0.7513	60,881.86
4	85,085.44	0.6830	58,114.50
5	89,339.71	0.6209	55,472.93
		NPV =	105,068.46

Decision:

Both methods yield the same NPV = Rs.105,068.46. This is higher than the NPV without inflation adjustment (Rs.65,355.07). The original analysis understated NPV by discounting real cash flows at the nominal rate. With proper inflation treatment, the project is even more attractive and **should be accepted**.

CHAPTER 12: RISK ANALYSIS IN CAPITAL BUDGETING

Question 1: Project Selection using Coefficient of Variation

Decision Rule: Compare using Coefficient of Variation ($CV = \text{Standard Deviation} / \text{Expected NCF}$). Lower CV implies less risk per unit of return.

(i)

Project	ENCF (Rs.)	Std Dev (Rs.)	CV = Sigma/ENCF
Project A	10,000	500	0.0500
Project B	10,000	1,000	0.1000

Same ENCF. Project A has lower risk (lower sigma). **Accept Project A.**

(ii)

Project	ENCF (Rs.)	Std Dev (Rs.)	CV = Sigma/ENCF
Project A	10,000	1,000	0.1000
Project B	12,000	500	0.0417

Project B has higher ENCF AND lower risk. B dominates A. **Accept Project B.**

(iii)

Project	ENCF (Rs.)	Std Dev (Rs.)	CV = Sigma/ENCF
Project A	500	500	1.0000
Project B	12,000	1,000	0.0833

B has much higher ENCF and much lower CV (0.08 vs 1.00). **Accept Project B.**

(iv)

Project	ENCF (Rs.)	Std Dev (Rs.)	CV = Sigma/ENCF
Project A	10,000	500	0.0500
Project B	12,000	600	0.0500

Both have same CV (0.05). B has higher ENCF. **Accept Project B.**

Question 3: Certainty Equivalent Approach - Walchand Company

Risk-Free Rate = 6%. Two mutually exclusive projects with certainty-equivalent coefficients.

Project A:

Year	Cash Flow	CE Coeff	Certain CF	PV Factor @6%	PV (Rs.)
0	-5,000	1.0	-5,000	1.0000	-5,000.00
1	1,000	0.9	900	0.9434	849.06
2	2,000	0.8	1,600	0.8900	1,423.99
3	3,300	0.7	2,310	0.8396	1,939.52
4	4,000	0.6	2,400	0.7921	1,901.02
5	1,000	0.3	300	0.7473	224.18
				NPV (A) =	1,337.77

Project B:

Year	Cash Flow	CE Coeff	Certain CF	PV Factor @6%	PV (Rs.)
0	-8,000	1.0	-8,000	1.0000	-8,000.00
1	6,000	0.9	5,400	0.9434	5,094.34
2	5,000	0.7	3,500	0.8900	3,114.99
3	4,000	0.6	2,400	0.8396	2,015.09
4	3,000	0.5	1,500	0.7921	1,188.14
5	1,000	0.25	250	0.7473	186.81
				NPV (B) =	3,599.37

Decision:

NPV(A) = Rs.1,337.77 and NPV(B) = Rs.3,599.37. **Select Project B** as it has a significantly higher NPV under the certainty-equivalent approach.

Risk-Adjusted Discount Rates (RADR):

Relationship: $\alpha_t / (1+R_f)^t = 1 / (1+RADR)^t$

Solving: $RADR = ((1+R_f) / \alpha_t^{1/t}) - 1$

Year	alpha (A)	RADR (A)	alpha (B)	RADR (B)
1	0.9	17.78%	0.9	17.78%
2	0.8	18.51%	0.7	26.69%
3	0.7	19.38%	0.6	25.68%
4	0.6	20.44%	0.5	26.06%
5	0.3	34.86%	0.25	39.87%

The RADR varies by year because the certainty-equivalent coefficients decline at different rates, reflecting increasing risk over time.

Case 12.1: Richa Foods Company

Investment = Rs.60 mn, Life = 6 years, 1 million units at Rs.60/unit, VC = Rs.30/unit, FC = Rs.20 mn (incl. depreciation), Tax = 35%, Required return = 12%.

Base Case NPV Calculation:

Particulars	Amount (Rs. mn)
Revenue (1 mn x Rs.60)	60.00
Variable Costs (1 mn x Rs.30)	30.00
Contribution	30.00
Fixed Costs (incl. depreciation)	20.00
Profit Before Tax	10.00
Tax @ 35%	3.50
Profit After Tax	6.50
Add: Depreciation (60/6)	10.00
Annual Cash Flow	16.50

PV Annuity Factor (6 yrs, 12%) = 4.1114

PV of Cash Flows = 16.50 x 4.1114 = Rs.67.84 mn

Base Case NPV = 67.84 - 60.00 = Rs.7.84 mn

Part 1: Sensitivity Analysis - Break-even Values (NPV = 0)

Required Annual CF for NPV = 0: Rs.60 / 4.1114 = Rs.14.59 mn

Required PAT = 14.59 - 10 (dep) = Rs.4.59 mn

Required PBT = 4.59 / 0.65 = Rs.7.07 mn

Break-even Parameter	Break-even Value	Base Case	Margin of Safety
Volume (mn units)	0.902	1.000	9.8%
Selling Price (Rs./unit)	57.07	60.00	4.9%
Variable Cost (Rs./unit)	32.93	30.00	9.8%
Investment (Rs. mn)	70.31	60.00	17.2%

The NPV is most sensitive to selling price (only 4.9% decline makes NPV zero) and volume (9.8% decline makes NPV zero). Variable cost and investment have larger margins of safety.

Part 2: Scenario Analysis

Parameter	Pessimistic	Expected	Optimistic
Investment (Rs. mn)	66	60	60
Volume (mn units)	0.95	1.00	1.10
Selling Price (Rs.)	54	60	69
Variable Cost (Rs.)	33	30	28.50

Particulars (Rs. mn)	Pessimistic	Expected	Optimistic
Revenue	51.30	60.00	75.90
Variable Costs	31.35	30.00	31.35
Contribution	19.95	30.00	44.55
Depreciation	11.00	10.00	10.00
PBT	-0.05	10.00	24.55
Tax @ 35%	-0.02	3.50	8.59
PAT	-0.03	6.50	15.96
Annual Cash Flow	10.97	16.50	25.96
PV of CFs	45.09	67.84	106.72

Particulars (Rs. mn)	Pessimistic	Expected	Optimistic
NPV	-20.91	7.84	46.72

Part 3: Recommendation:

The base case NPV is Rs.7.84 mn (positive). Under the optimistic scenario, NPV rises to Rs.46.72 mn. Under the pessimistic scenario, NPV falls to Rs.-20.91 mn.

Recommendation: The project should be accepted with caution. The base case is positive and the optimistic scenario shows strong upside. However, the pessimistic scenario shows a significant loss. The company should:

1. Conduct market research to estimate probabilities of each scenario
2. Analyse competitor response patterns and pricing behaviour
3. Consider a Monte Carlo simulation for a fuller risk picture
4. Explore strategies to reduce downside risk (flexible manufacturing, phased investment)